Dropout as a Bayesian Approximation

Representing Model Uncertainty in Deep Learning (Gal and Ghahramani, 2016)

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Probabilistic Machine Learning

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Presentation Overview

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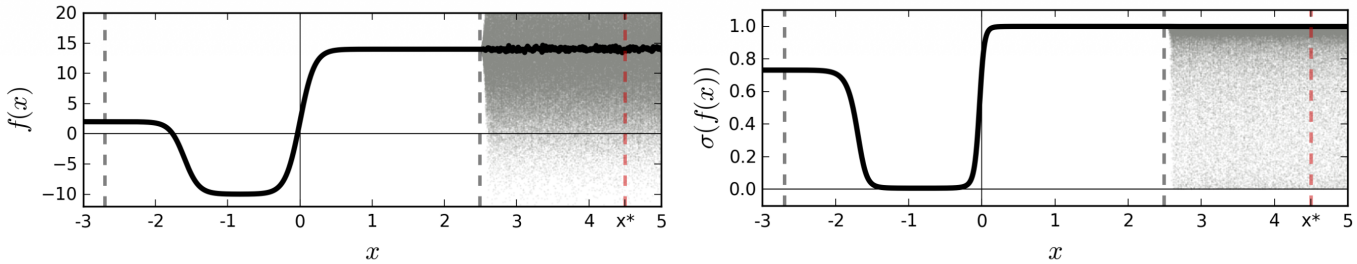
Overconfidence in Deep Networks



(a) Arbitrary function $f(\mathbf{x})$ as a function of data $\mathbf{x}$ (softmax *input*)

(b) $\sigma(f(\mathbf{x}))$ as a function of data $\mathbf{x}$ (softmax *output*)

Figure: A sketch of softmax input and output for an idealised binary classification problem

- Standard deep neural networks produce point estimates.
- Softmax outputs are often misinterpreted as calibrated confidence.
- Far from training data: predictions remain highly confident.
- Missing: **epistemic uncertainty** (model uncertainty).

Classical Dropout vs Bayesian Neural Networks

Bayesian Neural Networks

- Prior over weights: $p(\omega)$
- Posterior:

$$p(\omega|X, Y)$$

- Intractable for deep NNs
- MCMC (Gibbs, HMC) expensive
- This motivates approximate inference

Classical Dropout

- Randomly zero activations
- Prevents co-adaptation
- Interpreted as heuristic regularizer
- No explicit probabilistic view
- Can we view dropout as approximate Bayesian inference?

Dropout as Approximate Model Averaging

- Each dropout mask defines a sub-network.
- With K units per layer $\Rightarrow 2^K$ possible subnetworks.
- Training with dropout implicitly trains an ensemble of networks with shared weights.

Bayesian Perspective:
True predictive distribution:

$$p(y^*|x^*, X, Y) = \int p(y^*|x^*, \omega) p(\omega|X, Y) d\omega$$

Dropout approximation:

$$p(y^*|x^*) \approx \frac{1}{T} \sum_{t=1}^{T} p(y^*|x^*, \omega_t)$$

Dropout = Variational Inference

Variational Objective:

$$\mathcal{L}_{VI} = -\int q(\omega) \log p(Y|X,\omega) d\omega + \mathrm{KL}(q(\omega)||p(\omega))$$

- Dropout training $\equiv$ approximate VI in a deep Gaussian Process.
- Approximate posterior:

$$q(\omega)$$

- Bernoulli variational family:

$$W_i = M_i \cdot \mathrm{diag}(z_{i,j})$$

$$z_{i,j} \sim \mathrm{Bernoulli}(p_i)$$

Optimizing the ELBO

Standard Dropout Objective:

$$\mathcal{L}_{dropout} = \frac{1}{N}\sum_{i=1}^{N} E(y_i, \hat{y}_i) + \lambda \sum_{i=1}^{L}(||W_i||_2^2 + ||b_i||_2^2)$$

Dropout as Variational Approximation:

$$\mathcal{L}_{GP-MC} \propto \frac{1}{N}\sum_{n=1}^{N} \frac{-\log p(y_n \mid x_n, \hat{\omega_n})}{\tau} + \sum_{i=1}^{L}\left(\frac{p_i l^2}{2\tau N}||M_i||_2^2 + \frac{l^2}{2\tau N}||m_i||_2^2\right)$$

- L_2 regularization $\leftrightarrow$ Gaussian prior
- Training minimizes KL divergence to posterior
- Equivalent to maximizing the ELBO
- Standard dropout loss is equivalent to the negative ELBO

Connection to Deep Gaussian Processes

- Single-layer NN with infinite width $\Rightarrow$ Gaussian Process.
- Dropout introduces multiplicative Bernoulli noise.
- This induces a variational approximation to a Deep GP posterior.

$$f^{(l)}(x) = W^{(l)}\sigma(f^{(l-1)}(x))$$

Under dropout:

$$W^{(l)} = M^{(l)} \cdot \mathrm{diag}(z)$$

- Random masks approximate integrating over function space.

Monte Carlo Dropout

- Keep dropout active at test time.
- Perform T stochastic forward passes and average them.
- Approximate intractable posterior predictive distribution.

Predictive Mean:

$$\mathbb{E}[y^*] \approx \frac{1}{T} \sum_{t=1}^{T} \hat{y}^{*t}$$

Predictive Variance:

$$\mathrm{Var}(y^*) \approx \tau^{-1} I + \frac{1}{T} \sum_{t=1}^{T} \hat{y}^{*t} \hat{y}^{*tT} - \mathbb{E}[y^*]\mathbb{E}[y^*]^T$$

- Variance term captures epistemic uncertainty.
- $\tau^{-1} I$ captures model precision (linked to prior).
- Uncertainty increases far from training data.

Predictive Log-Likelihood and RMSE Comparison

	Avg. Test RMSE and Std. Errors			Avg. Test LL and Std. Errors		
Dataset	**VI**	**PBP**	**Dropout**	**VI**	**PBP**	**Dropout**
Boston Housing	4.32 ±0.29	3.01 ±0.18	**2.97 ±0.85**	-2.90 ±0.07	-2.57 ±0.09	**-2.46 ±0.25**
Concrete Strength	7.19 ±0.12	5.67 ±0.09	**5.23 ±0.53**	-3.39 ±0.02	-3.16 ±0.02	**-3.04 ±0.09**
Energy Efficiency	2.65 ±0.08	1.80 ±0.05	**1.66 ±0.19**	-2.39 ±0.03	-2.04 ±0.02	**-1.99 ±0.09**
Kin8nm	**0.10 ±0.00**	**0.10 ±0.00**	**0.10 ±0.00**	0.90 ±0.01	0.90 ±0.01	**0.95 ±0.03**
Naval Propulsion	**0.01 ±0.00**	**0.01 ±0.00**	**0.01 ±0.00**	3.73 ±0.12	3.73 ±0.01	**3.80 ±0.05**
Power Plant	4.33 ±0.04	4.12 ±0.03	**4.02 ±0.18**	-2.89 ±0.01	-2.84 ±0.01	**-2.80 ±0.05**
Protein Structure	4.84 ±0.03	4.73 ±0.01	**4.36 ±0.04**	-2.99 ±0.01	-2.97 ±0.00	**-2.89 ±0.01**
Wine Quality Red	0.65 ±0.01	0.64 ±0.01	**0.62 ±0.04**	-0.98 ±0.01	-0.97 ±0.01	**-0.93 ±0.06**
Yacht Hydrodynamics	6.89 ±0.67	**1.02 ±0.05**	1.11 ±0.38	-3.43 ±0.16	-1.63 ±0.02	**-1.55 ±0.12**
Year Prediction MSD	9.034 ±NA	8.879 ±NA	**8.849 ±NA**	-3.622 ±NA	-3.603 ±NA	**-3.588 ±NA**

Figure: Predictive performance (RMSE and Log-Likelihood) across VI, PBP, and MC Dropout. MC Dropout achieves competitive RMSE and higher log-likelihood, showing better uncertainty estimation.

- Demonstrates principled Bayesian behavior.
- RMSE shows accuracy, Log-Likelihood shows calibrated uncertainty.
- BO (Bayesian Optimization) further improves performance.

MC Dropout in Action

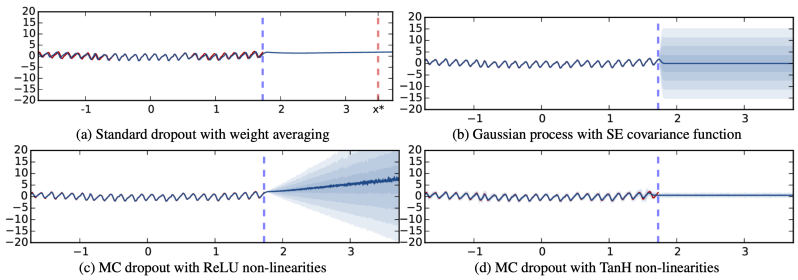


Figure: Regression: predictive mean $\pm$ uncertainty.

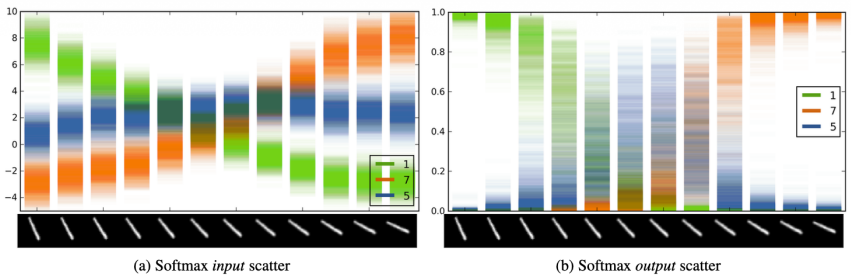


Figure: Classification: softmax uncertainty on rotated digits.

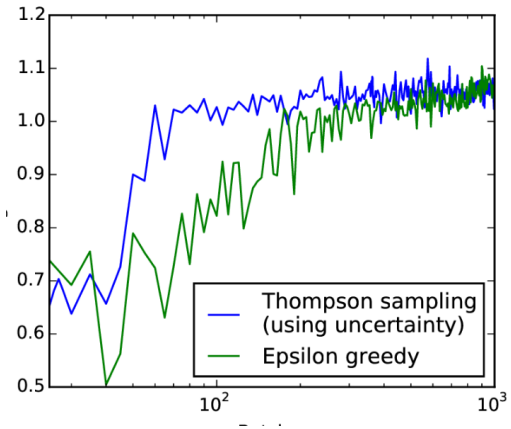


Figure: RL: MC Dropout (Thompson sampling) vs ϵ-greedy average reward.

Why MC Dropout is Powerful

Dropout casts Bayesian inference as a fast optimization problem rather than a sampling problem.

- Uncertainty estimates using existing models.
- No doubled parameterization.
- No burn-in.
- Scales to modern deep architectures.

Trade-off:

- VI: fast, biased (mode-seeking).
- MCMC: accurate, computationally heavy.

Conclusion and Limitations

Conclusion

- Dropout reinterpreted as approximate Bayesian inference.
- Provides fast, scalable uncertainty estimation.

Limitations

- Variational family is mean-field Bernoulli: cannot represent multimodal posteriors.
- $\mathrm{KL}(q||p)$ is mode-seeking: underestimates posterior variance.
- Prior is implicitly tied to weight decay coefficient.
- Approximation quality depends on dropout rate choice.

References

Yarin Gal and Zoubin Ghahramani (2016)
Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning